



Design Document

Product: MarkForge (Premium Markdown Studio)

1. Vision

Build a minimal, elite Markdown workspace that feels like:

VS Code × Notion × Linear — but focused purely on Markdown.

Core goals:

- Elite editor experience
 - Beautiful live preview
 - Professional PDF export
 - File system integration
 - Multi-document workspace
 - Zero clutter, maximum focus
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2. Core Feature Set (Expanded)

2.1 Core Writing Experience

- Split view: Editor + Preview
 - Real-time Markdown rendering
 - Dark / Light mode
 - Download as high-quality PDF
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2.2 File System Integration (NEW)

A. Open Single File

User should be able to:

- Click “Open File”
- Select `.md` file from local directory
- Load content into editor

OR

- Drag & drop `.md` file directly into the workspace
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B. Drag & Drop Folder (Advanced)

User can:

- Upload a folder
- System scans recursively
- Detect all `.md` files
- Display them in a sidebar (VS Code-style structure)

Implementation:

- Use File System Access API (for supported browsers)
 - Fallback to `<input type="file" webkitdirectory>`
-

C. Folder Tree Sidebar

Left sidebar appears only when folder is loaded.

Structure:

| Sidebar (File Tree) | Editor | Preview |

Features:

- Collapsible folders
- Highlight active file
- File rename (optional future)
- Create new file (future phase)
- Delete file (future phase)

UI Inspiration:

- VS Code minimal file explorer
 - Clean indentation
 - No heavy borders
 - Hover reveal icons
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3. Multiple Editor System

3.1 Tabs (Multi-Document)

Top of editor area shows tabs:

| file1.md | notes.md | README.md | + |

Features:

- Multiple open files
 - Switch instantly
 - Close tab
 - Unsaved indicator (●)
 - Keyboard shortcut: ⌘ + P (Quick file switcher)
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3.2 State Management

Each file maintains:

- Content
- Scroll position
- Cursor position
- Dirty state

When switching tabs:

- Preserve editor state
 - Preserve preview scroll
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4. Export System (Expanded)

4.1 Export Current File

Download:

- PDF (via Puppeteer)
 - Raw `.md` (optional button)
-

4.2 Export Entire Workspace (NEW)

If folder loaded:

User can click:
"Export Project"

Options:

- Export as ZIP
 - Export as merged PDF
 - Export as individual PDFs in ZIP
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ZIP Export Logic

When exporting ZIP:

If multiple files:

- Preserve folder structure
- Include all `.md` files
- Optionally include generated PDFs

Use:

- JSZip (frontend)
OR
 - Backend ZIP generation for larger projects
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5. Layout (Updated)

5.1 Base Layout (No Folder Loaded)

Top Command Bar
Editor (50%)

5.2 Folder Mode Layout

Top Command Bar
Sidebar

Sidebar width:

- 240px
- Collapsible
- Resizable

6. UI System (Refined)

6.1 Sidebar Design

Light Mode:

- Background: #F3F4F6

Dark Mode:

- Background: #111827

Visual Rules:

- Subtle indentation

- Hover highlight
 - Active file accent line (left border)
 - Smooth expand/collapse animation
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6.2 Tabs Design

- Height: 40px
 - Rounded corners (top)
 - Soft inactive background
 - Active tab elevated slightly
 - Close icon on hover
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7. Drag & Drop UX

7.1 Single File Drop

When dragging:

- Show overlay
- "Drop Markdown File"

After drop:

- Auto-load into editor
 - Add tab
-

7.2 Folder Drop

When folder dropped:

- Show scanning state
 - Parse all `.md`
 - Build file tree
 - Open first file automatically
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8. Technical Architecture (Updated)

Frontend:

- React + Vite
- CodeMirror 6
- markdown-it
- Tailwind CSS
- Zustand (state)
- JSZip

Backend:

- Node + Express
 - Puppeteer (PDF generation)
 - Optional backend ZIP for large projects
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9. Data Model

File Object

```
{
  id: string,
  name: string,
  path: string,
  content: string,
  isDirty: boolean,
  cursorPosition: object,
  scrollPosition: number
}
```

Folder Tree Structure

```
{
  name: string,
  type: "folder" | "file",
```

```
children?: []  
}
```

10. Performance Targets

Metric	Target
Folder scan (100 files)	< 500ms
File switch delay	< 30ms
ZIP export (10 files)	< 2s
PDF export	< 3s

11. Edge Cases

- Very large `.md` file (>1MB)
 - Deep nested folder structure
 - Non-markdown files (ignore)
 - Duplicate file names
 - Special characters in file names
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12. Security Considerations

- Sanitize Markdown before rendering
 - Prevent script injection
 - Restrict file access to user-selected directory only
 - No automatic background file reads
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13. Competitive Advantage After These Additions

Now this tool becomes:

Not just a Markdown editor.

But a:

- Lightweight Markdown IDE
- Documentation workspace
- Project-based writing system

Competitors:

- Single document tools
- No folder structure
- Weak export capabilities

MarkForge:

- Project-level organization
 - Multi-tab editing
 - Professional exports
 - Premium UI
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14. Real-World Use Case

You create a folder:

STQA-Notes/

Unit1.md

Unit2.md

Unit3.md

You drag entire folder.

Sidebar shows structure.

You edit files.

When done:

Click "Export Project"

→ Download ZIP containing:

- All `.md`

- Or compiled PDFs

Clean. Structured. Efficient.

15. TLDR (Expanded Scope)

MarkForge now includes:

- Editor + Preview
- Open file from directory
- Drag & drop file
- Upload full folder
- Sidebar file tree
- Multi-tab editing
- Export single file as PDF
- Export full project as ZIP

This transforms it from a utility tool into a lightweight Markdown IDE.

Premium UI.

Professional workflow.

No compromise.